

Comparing Means: T-Test

BIOS227 | PSYC227 | ENVS227

Recap: Hypothesis Testing

1. Define Null and Alternative Hypothesis
2. Set a fixed Type 1 error rate you are 'okay' with.
3. Based on different assumptions about the data, use a test-statistic

Note: If the null hypothesis holds, this test statistic will follow a fixed distribution

4. Calculate test statistic for given Data
5. Check how 'extreme' the test statistic value is. This can be done by:
 - a. Comparing against critical test-statistic value
 - b. Calculate p-value
6. Decide whether to accept or reject the null hypothesis based on how 'extreme' the test statistic is with respect to the Type 1 error rate

Recap: Z-test

Z-test is used to compare means when:

- Population Variance is known
- But this is rarely ever the case
 - We can never really know the exact variance in human heights across the population, but we can get a good estimate

A t-test is similar to the Z-test in that it also compares means, *but it does so when population-wide variance σ^2 is not available.*

Note: Many of the graphs and intuitions I cover are not limited to the t-test and extrapolate well to the Z-test.

T-Test

We use the T-test when:

- We need to compare means – Some sub-types:
 - **One Sample T-test** (One mean vs given Value): Say a company claims that it gives 200g of pesticide per unit of product. You want to check if you are being cheated by looking at a sample
 - **Two-Sample T-test** (Mean vs Mean): To see if a Plant species A requires more pesticide than Species B for healthy growth
 - **Paired T-test** (Past vs Future Mean): If a Plant growth rate differs before and after pesticide usage
- Population variance unknown: Here, we use sample SD as a proxy for σ
- Usually, when sample size $n \leq 30$
 - Useful when samples could be hard to come-by (say while studying rare diseases)
 - Sometimes, Z-test used when $n \geq 30$, but t-test is considered more robust

Recap: Hypothesis Testing

1. Define Null and Alternative Hypothesis
2. Set a fixed Type 1 error rate you are 'okay' with.
3. Based on different assumptions about the data, use a test-statistic

Note: If the null hypothesis holds, this test statistic will follow a fixed distribution. **We will cover assumptions for a t-test a bit later**

- 4. Calculate test statistic for given Data**

The 't' statistic

One-Sample t-test

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

Two-Sample/Paired t-test

$$t = \frac{(\bar{x}_1 - \bar{x}_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

The t-statistic is a *signal-to-noise ratio*: How big is the mean difference compared to its standard error

An intuition for the t-statistic

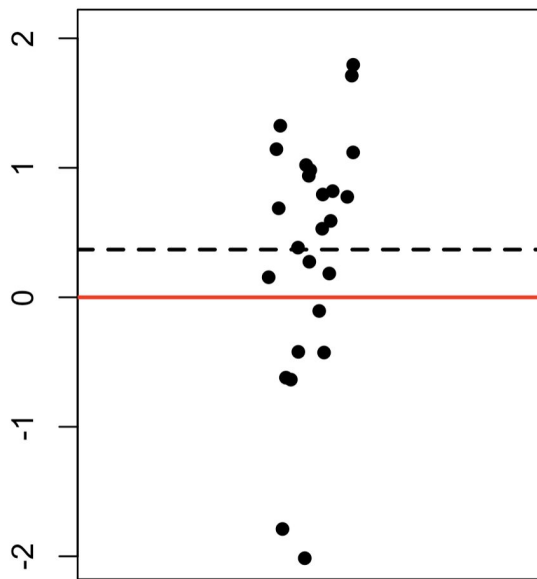
For simplicity, we consider the test statistic for a one sample t-test, i.e., $t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$

Consider these two graphs:

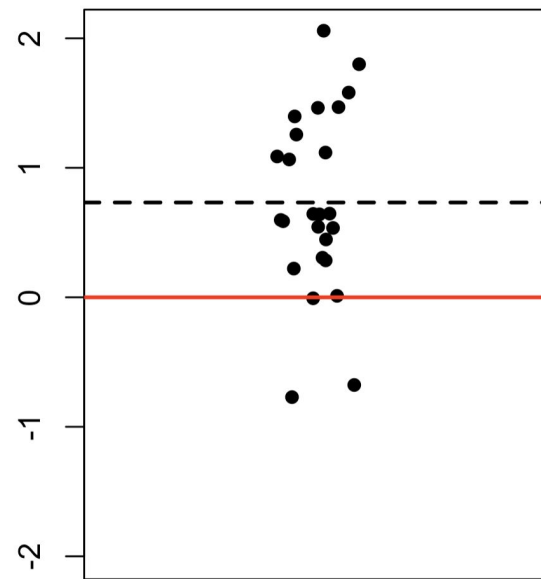
Note: Red line represents μ ,
Dotted line is the sample mean $\bar{x}$,
standard deviation for both
samples is s

Is the t -statistic for Sample A
greater than or *lesser than* the
 t -statistic for Sample B?

Sample A



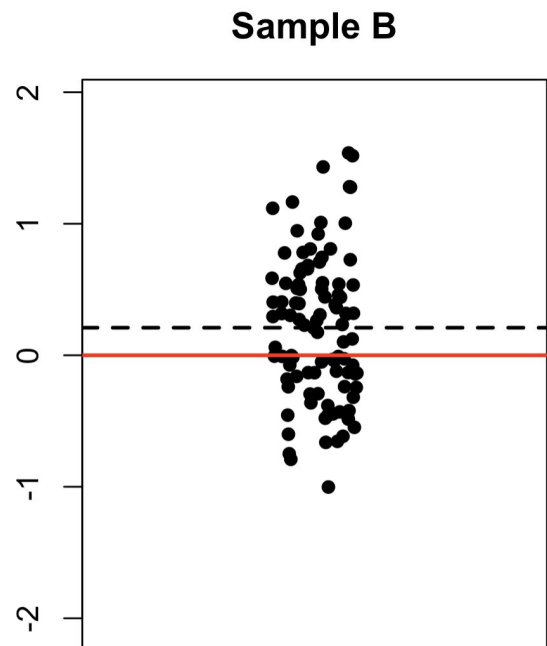
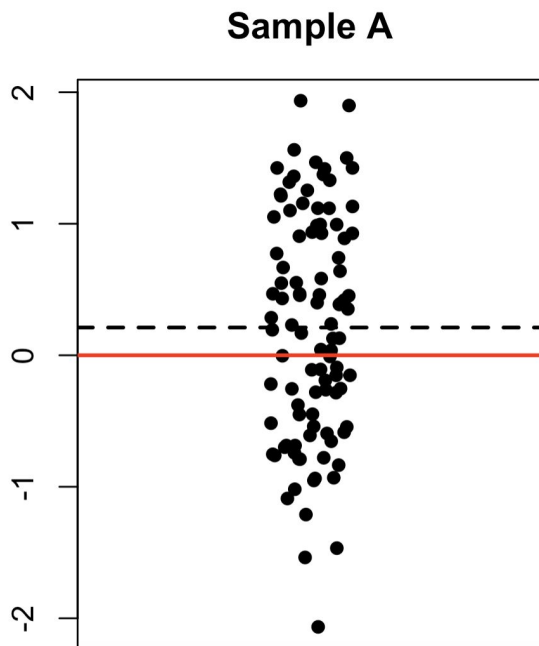
Sample B



An intuition for the t-statistic

Now, consider these two graphs: Which sample has the greater t-statistic?

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$



t-statistic

Thus,

- **Difference in means grows** → numerator grows → **|t| grows.**
- **Variance grows** → standard error grows → **|t| shrinks.**

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

Exercise: Try and visualise what this visualisation schema would look like for a two-sample t-test

A few takeaways:

- Same mean difference looks “more convincing” when variability is low.
- Same variability looks “more convincing” when the mean difference is huge.
 - If two groups’ sample means differ by 1 unit: with SD=0.1 that’s huge; with SD=10 that’s negligible

The Types of T-Test

- **One Sample T-test** (One mean vs given Value):

Say a company claims that it gives 200g of pesticide per unit of product and you want to check if you are being cheated by looking at a sample

- **Two-Sample T-test** (Mean vs Mean):

To see if a Plant A requires more pesticide than Plant B for healthy growth

- **Paired T-test** ('Past' Mean vs 'Future' Mean):

Is there a difference in growth rate of a plant before and after the usage of pesticide?

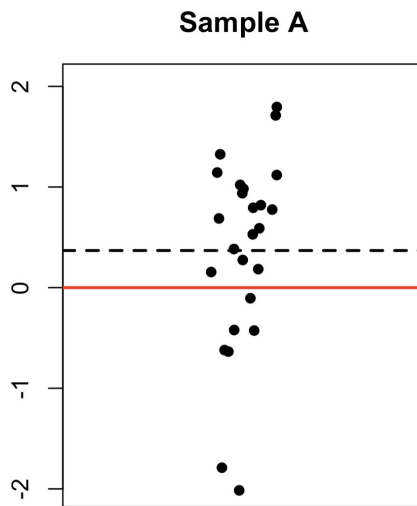
One Sample T-test

A One-sample T-test is used to compare a sample with mean ($\bar{x}$) and standard deviation (s) to a fixed reference mean (μ).

Null Hypothesis: $\bar{x} = \mu$

Alternative Hypothesis:

- Two-tailed: $\bar{x} \neq \mu$
- One-tailed:
 - $\bar{x} > \mu$
 - $\bar{x} < \mu$



test- statistic

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

Degrees of Freedom: $n - 1$, where n is the sample size

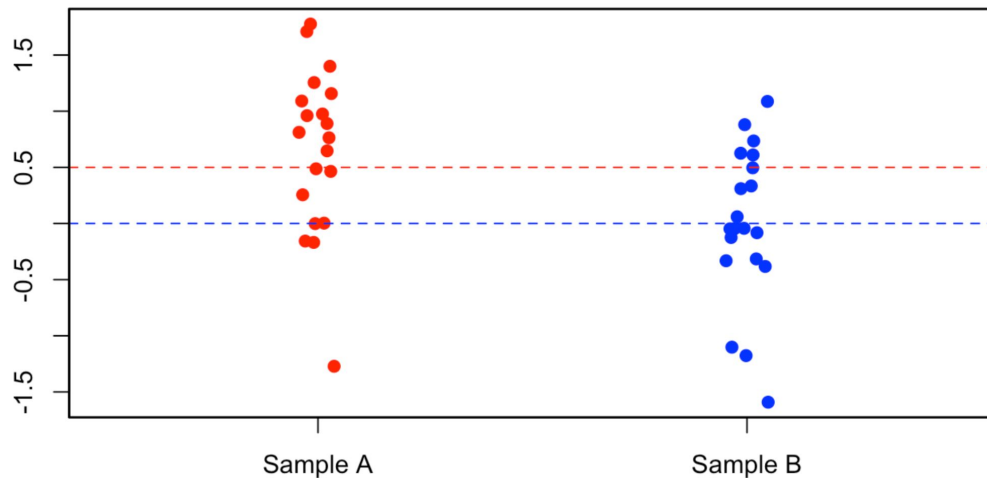
Two Sample T-test

A Two-sample T-test is used to compare the mean of sample 1 ($\bar{x}_1$) to the mean of sample 2 ($\bar{x}_2$)

Null Hypothesis: $\bar{x}_1 = \bar{x}_2$

Alternative Hypothesis:

- Two-tailed: $\bar{x}_1 \neq \bar{x}_2$
- One-tailed:
 - $\bar{x}_1 < \bar{x}_2$
 - $\bar{x}_1 > \bar{x}_2$



Degrees of Freedom: $n_1 + n_2 - 2$, where n_i is the sample size of sample i

Two Sample T-test

Here, there are 2 possibilities

- Samples 1 and 2 have unequal variance s_1 and $s_2 \Rightarrow$ Welch t-test
- Samples 1 and 2 have equal variance s

We use Levene's test to see if variance is equal

test- statistic
(Unequal variance)

$$t = \frac{(\bar{X}_1 - \bar{X}_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

test- statistic
(Equal variance)

$$t = \frac{(\bar{X}_1 - \bar{X}_2)}{\sqrt{\frac{s^2}{n_1} + \frac{s^2}{n_2}}}$$

Two Sample T-test

An Alternative (and more general) two-sample t-statistic:

$$\mathbf{t} = \frac{(\overline{x_1} - \overline{x_2}) - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

Allows us to compute for more general hypothesis:

- Alternative Hypothesis: The difference between means is greater than 5 units, i.e., $\bar{x}_1 - \bar{x}_2 > 5$

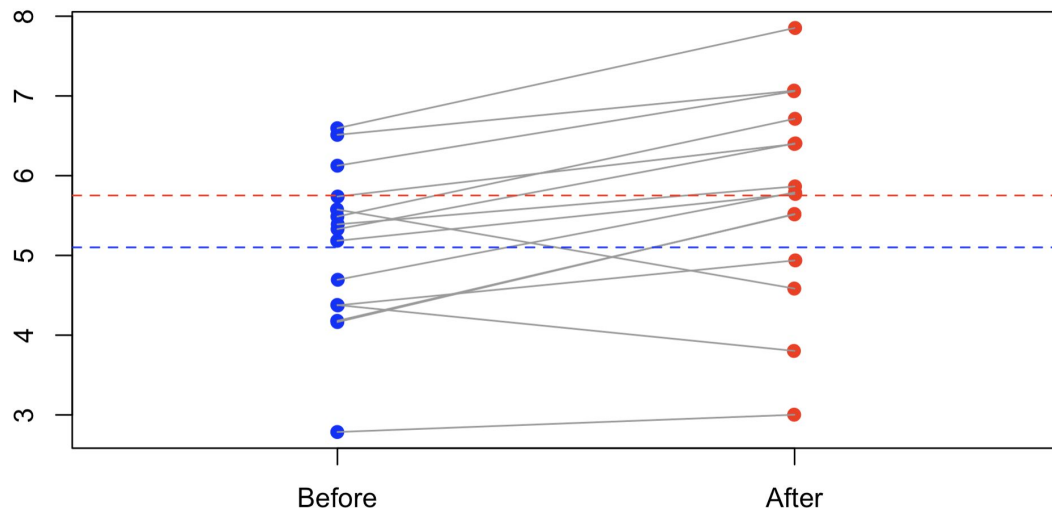
Paired T-test

A Paired sample T-test is used to compare the mean of a variable drawn from a sample before a 'treatment' ($\bar{x}_{\text{before}}$) and the mean after the 'treatment' ($\bar{x}_{\text{after}}$), i.e., there is temporal separation.

Null Hypothesis: $\bar{x}_{\text{before}} = \bar{x}_{\text{after}}$

Alternative Hypothesis:

- Two-tailed: $\bar{x}_{\text{before}} \neq \bar{x}_{\text{after}}$
- One-tailed:
 - $\bar{x}_{\text{before}} < \bar{x}_{\text{after}}$
 - $\bar{x}_{\text{before}} > \bar{x}_{\text{after}}$



Degrees of Freedom: $n - 1$, where n is the sample size

Sample Size ≤ 30 ?

No

Use Z-test

Yes

How many Samples?

Two Samples

One-Sample

Use One-Sample t-test

Are they independent?

No

Use Paired t-test

Yes

Two-Sample t-test

Equal Sample Variance
 $\Rightarrow$ **Pooled Two-sample**

Unequal Sample Variance
 $\Rightarrow$ **Welch t-test**

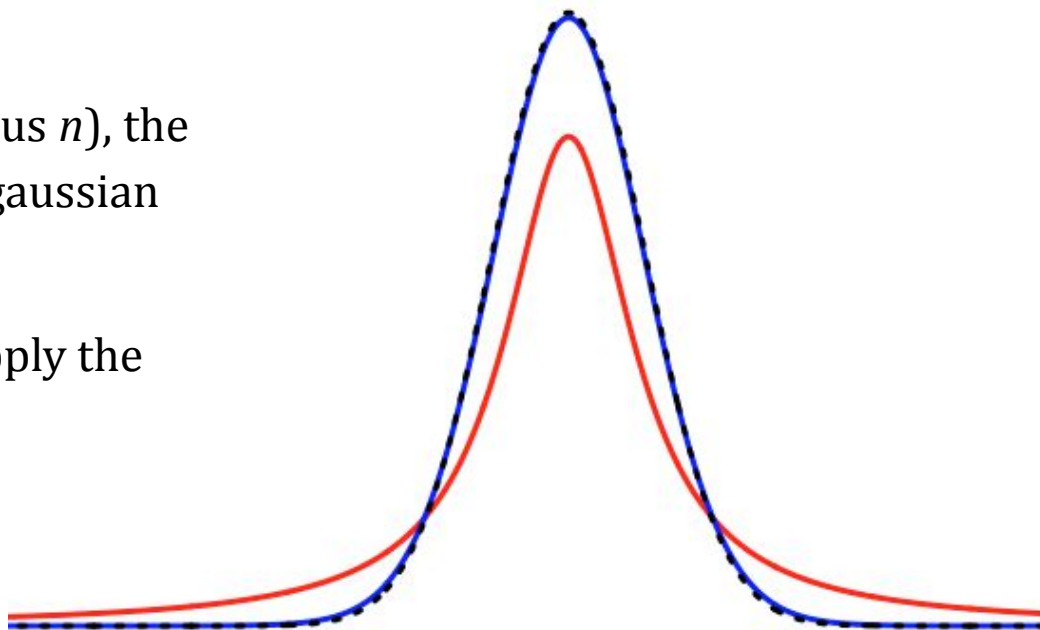
Degrees of Freedom: Intuition

This parameter controls the *Thickness of the tails* (i.e., kurtosis)

With larger degrees of freedom (and thus n), the t-distribution roughly approaches the gaussian distribution

- This is where the ' $n \geq 30$ rule' to apply the z-test instead of t-test comes from

— t (df = 1)
— t (df = 30)
- - - Normal $N(0,1)$



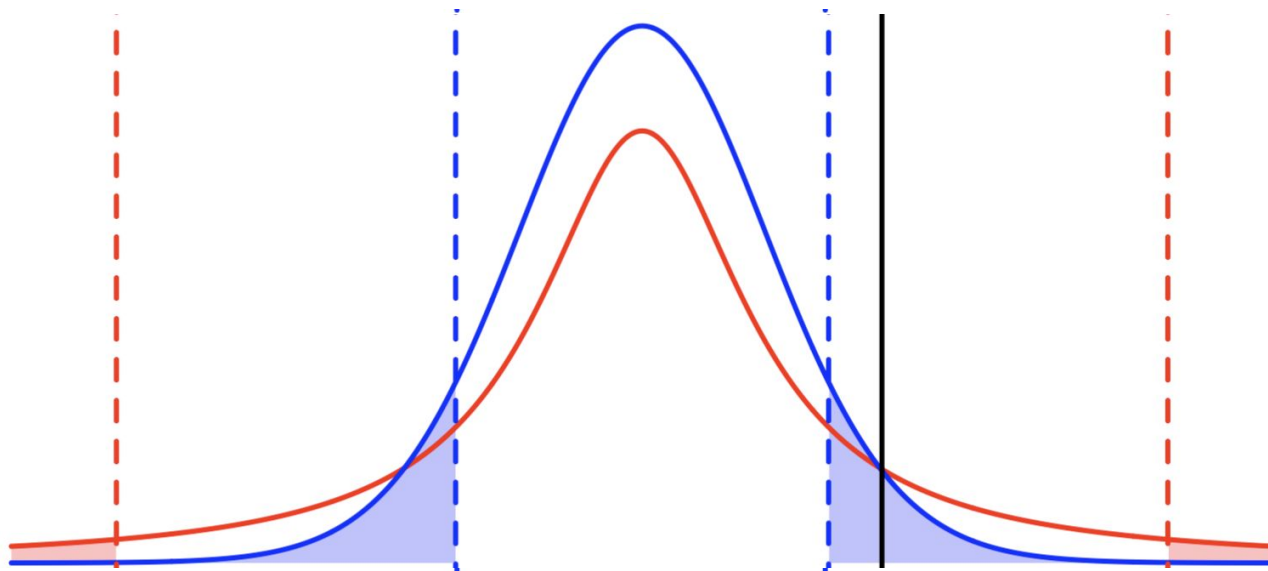
Degrees of Freedom: Modelling Uncertainty

- Roughly, the degrees of freedom parameter models uncertainty. There is a more formal mathematical definition, but not *very* important
- When you have a low sample size, intuitively, it should be more difficult to reject the null hypothesis with a smaller sample size, i.e., *you need 'greater extremeness' of the test statistic*
 - i.e., we'd want to be more certain that a difference exists before rejecting the null

Basically, a smaller sample $\Rightarrow$ noisier estimate $\Rightarrow$ we need to be more cautious before claiming a real difference

Degrees of Freedom: Modelling Uncertainty

— t (df = 1)
— t (df = 30)



For the same test statistic value (in black), if we have a greater sample size, we can reject the null (**blue rejection region**). We can't reject the null if we have a lower degree of freedom (**in red**) as we are more uncertain about the the result.

Degrees of Freedom: For different tests

One-Sample: $n - 1$ $\mathbf{t} = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$

Two-Sample: $n_1 + n_2 - 2$ $\mathbf{t} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$

Paired: $n - 1$ $\mathbf{t} = \frac{(\bar{X}_1 - \bar{X}_2)}{\frac{s}{\sqrt{n}}}$

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4. Calculate test statistic for given Data
- 5. Check how 'extreme' the test statistic value is. This can be done by:**
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 - b. Calculate p-value**
- 6. Decide whether to accept or reject the null hypothesis based on how 'extreme' the test statistic is with respect to the Type 1 error rate**

Recap: One-tailed and Two-tailed test

Two-tailed:

- You care about deviations in **either direction**.
- p-value counts extremeness on both sides.

One-tailed:

- **Directional claim** decided before seeing data.
- p-value counts extremeness in one direction only.

Based on Alternative hypothesis Statement, we choose whether we do a one-tailed or two-tailed test

Testing 'Extremeness' of Observation

A good interactive resource to 'test extremeness':

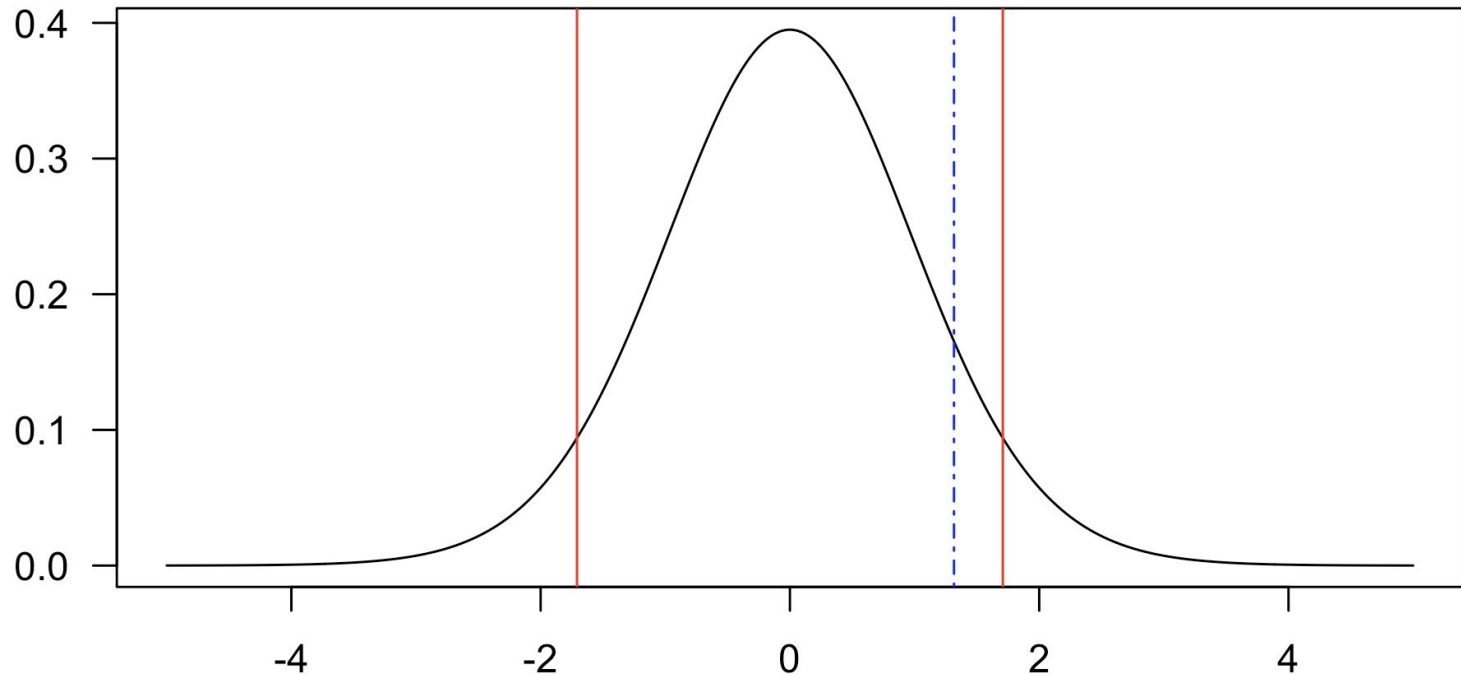
<https://observablehq.com/@bluprince13/an-interactive-explanation-of-the-t-test-to-compare-means>

There are Two Equivalent ways to test 'extremeness':

- Using the **test statistic**:
 - Calculate critical t-value (in R using the qt function, or t-tables)
 - Compare observed test-statistic and critical t-value
- Using the **p-value**:
 - Calculate the p-value of observation
 - Compare with Type-I Error Rate

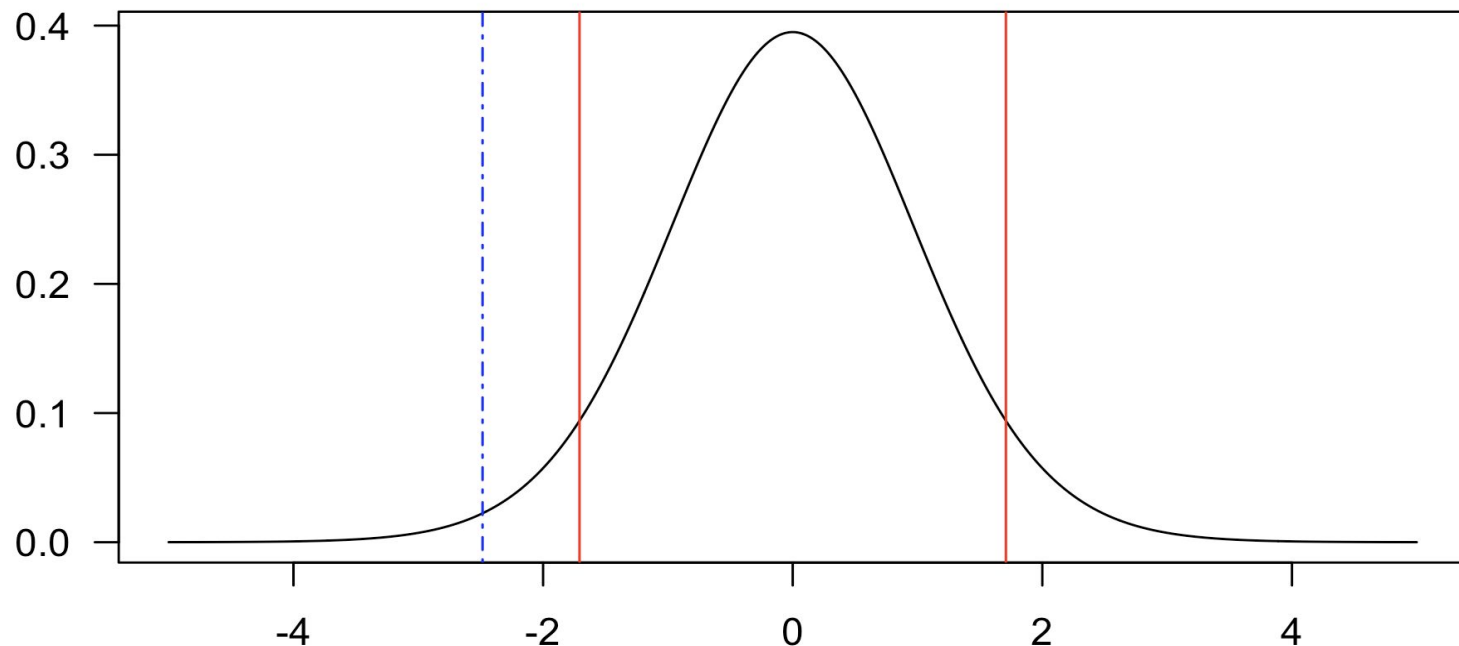
Testing Extremeness: Using test statistic

- Two-Tailed: **Accept** or **Reject** the Null?



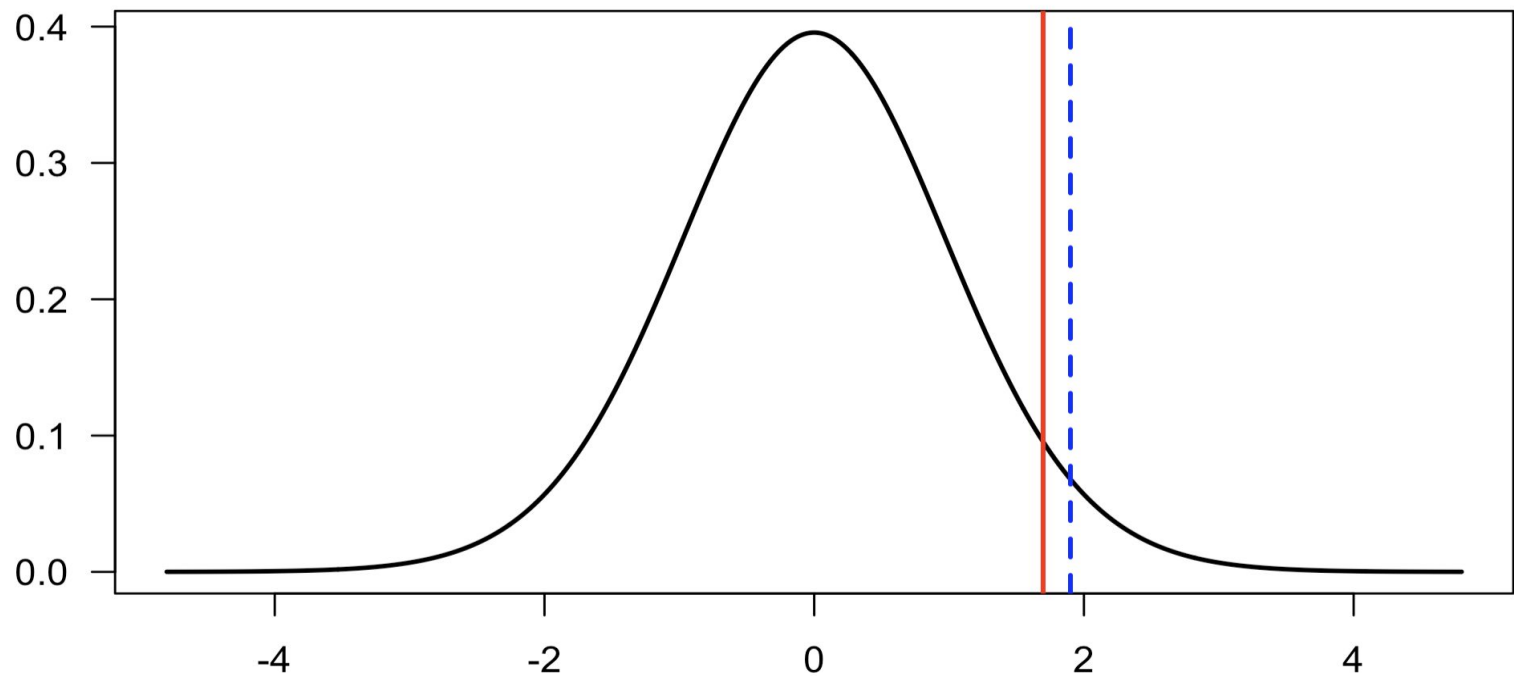
Testing Extremeness: Using test statistic

- Two-Tailed: **Accept** or **Reject** the Null?



Testing Extremeness: Using test statistic

- One-Tailed: **Accept** or **Reject** the Null?



Testing Extremeness: Using p-value

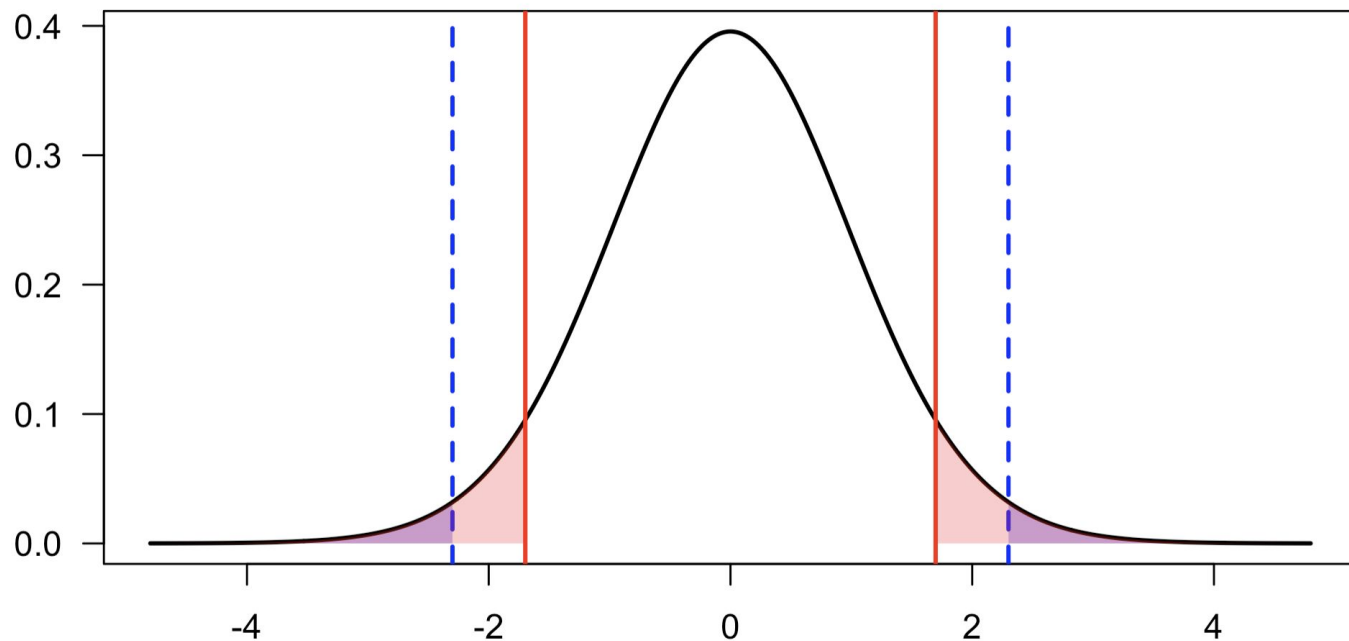
Accept or **Reject** the Null?

For Two-Tailed test:

Given, $\alpha = 0.1$

From observation,

$p = 0.05$



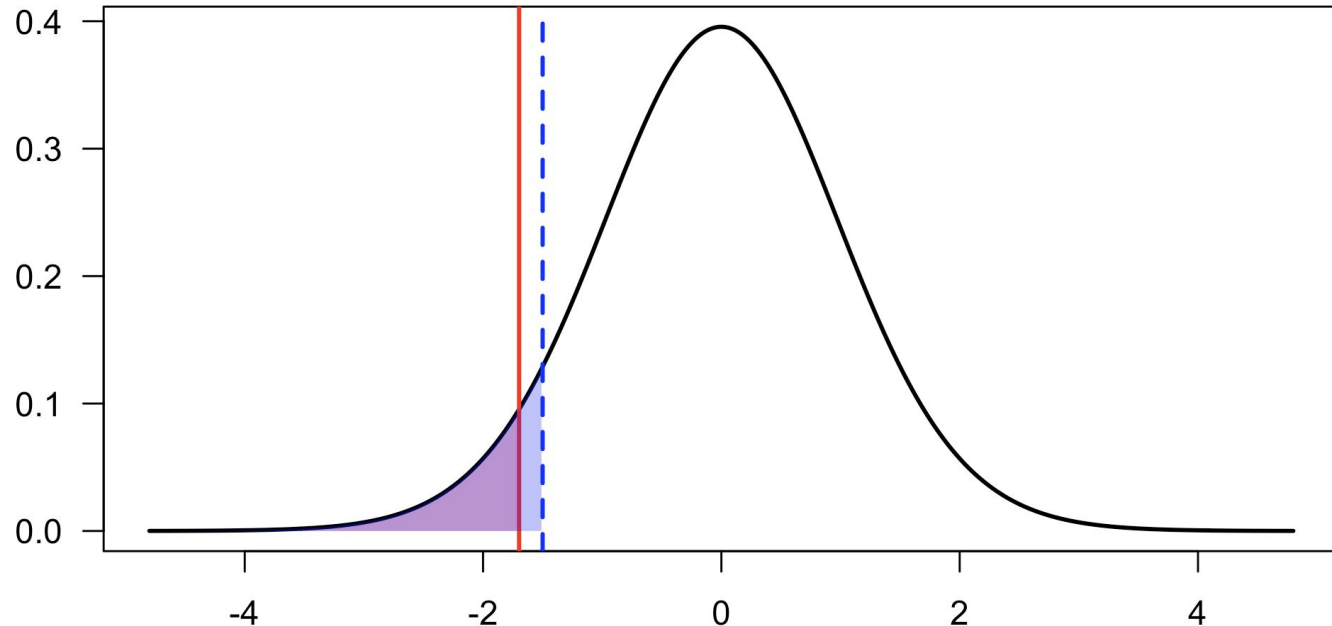
Testing Extremeness: Using p-value

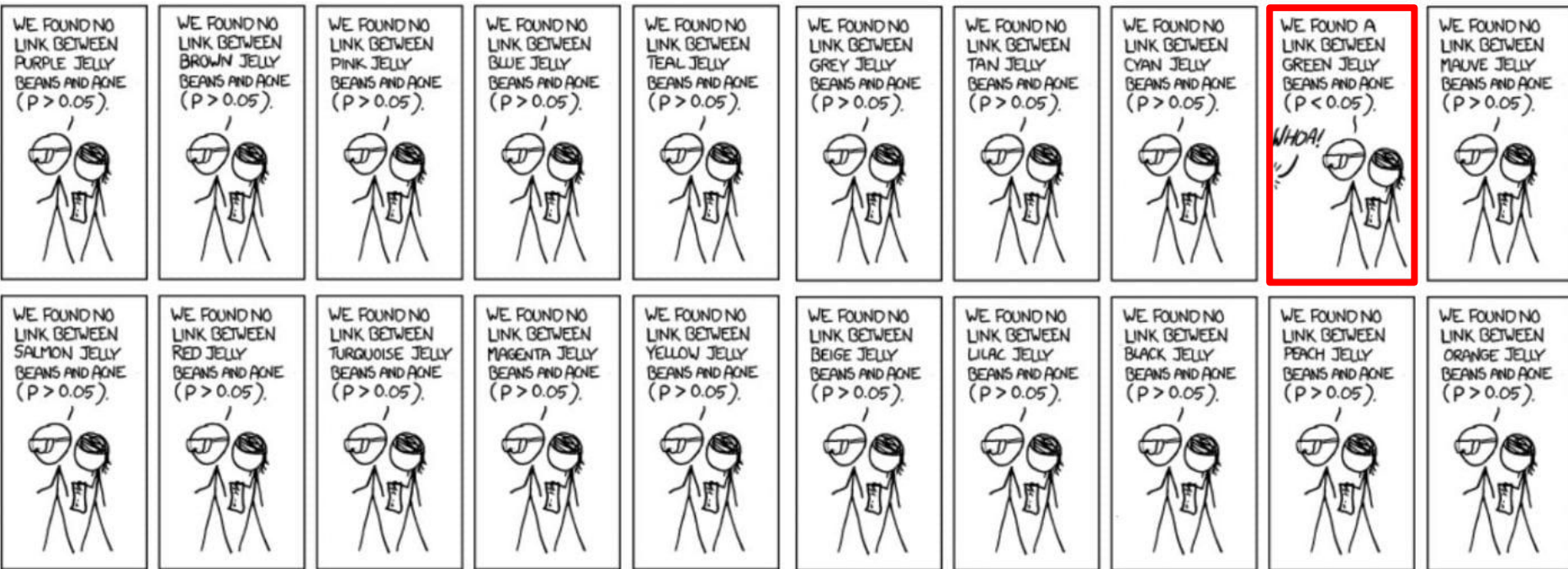
Accept or **Reject** the Null?

For One-Tailed test:

Given, $\alpha = 0.05$

From observation,
 $p = 0.06$





Assumptions:

This should be done before the test is applied. We cover it later for pedagogical purposes:

Core assumptions:

- **Approximately normal residuals** (or normality of differences for paired).
- For pooled two-sample: **equal sample variances**.

Note: t-tests are robust to **mild non-normality**, especially with moderate n and balanced group sizes.

Checking Assumptions: Normality

Normality of What???

- **One-sample**: the residuals
- **Two-sample**: each group distribution (or residuals)
- **Paired**: the differences between pairs

Checking Assumptions: Normality

How do we check normality?

- Plotting the Distribution: Qualitative
- Q-Q plot: look for severe departures / heavy tails.
- Shapiro–Wilk: Stricter especially with large n

Note: t-test is quite robust to mild normality, which is why we prefer the Q-Q Plots

Checking Assumptions: Normality

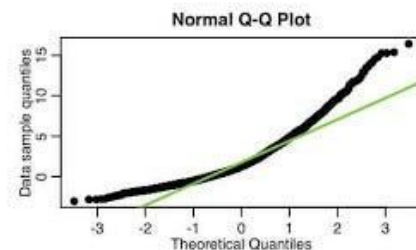
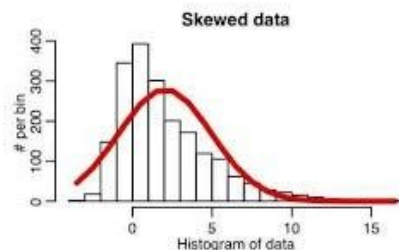
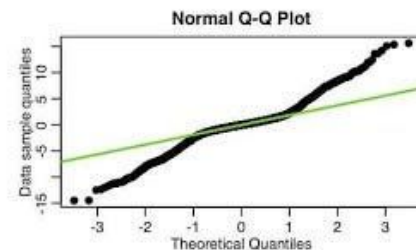
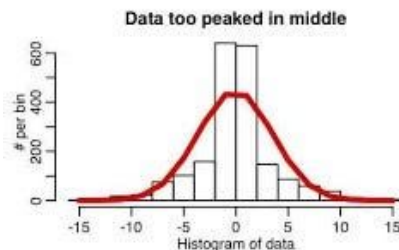
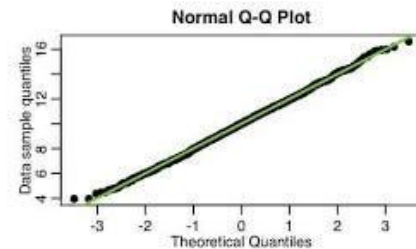
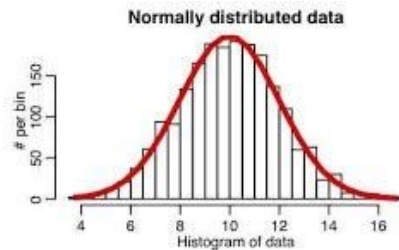
The Q-Q plot: This plot compares the **Theoretical Quantiles from a Normal Distribution** and **Actual Quantiles**

Resource:

<https://www.youtube.com/watch?v=X9 ISJ0YpGw>

*If the theoretical distribution aligns with the actual distribution, there will be **no deviations from the diagonal***

The Shapiro-Wilk test uses deviations from the diagonal to formally test for normality



Checking Assumptions: Variance

- Equal Variance: If we can assume equal variances
 - **Rule of Thumb:** $0.5 < \text{Variance (Group 1)} / \text{Variance (Group 2)} < 2$
 - **Levene's test:** Formal test to check for equal variances

R-Walkthrough: The t.test function

The 'from-scratch' implementation not shown here: Such an implementation of a two-sample t-test is present in the RMD file.

The t.test function: <https://www.rdocumentation.org/packages/stats/versions/3.6.2/topics/t.test>

```
t.test(x, y = NULL,  
       alternative = c("two.sided", "less", "greater"),  
       mu = 0, paired = FALSE, var.equal = FALSE,  
       conf.level = 0.95, ...)
```

The t.test function in R

```
t.test(x, y = NULL,  
       alternative = c("two.sided", "less", "greater"),  
       mu = 0, paired = FALSE, var.equal = FALSE,  
       conf.level = 0.95, ...)
```

x: a (non-empty) numeric vector of (Continuous) data values

y: an optional (non-empty) numeric vector of (Continuous) data values – Used for two-sample t-test

alternative: a string specifying the alternative hypothesis. It must be one of "two.sided" (default), "greater" or "less".

mu: a number indicating the true value of the mean (or difference in means if you are performing a two sample test).

The t.test function in R

```
t.test(x, y = NULL,  
       alternative = c("two.sided", "less", "greater"),  
       mu = 0, paired = FALSE, var.equal = FALSE,  
       conf.level = 0.95, ...)
```

paired: a logical variable (TRUE/FALSE) indicating whether you want a paired t-test.

var.equal: a logical variable indicating whether to treat the two variances as being equal.

conf.level: confidence level of the interval, i.e., $1 - \alpha$

How to use the t.test function: **One Sample t-test**

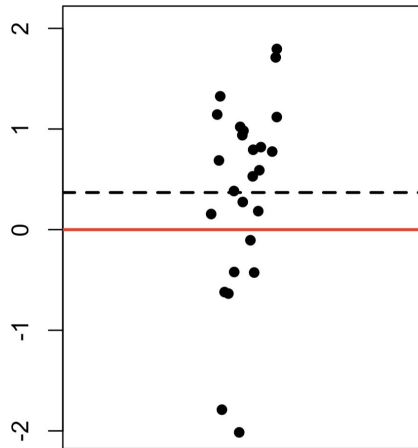
Example: A study claims that people with a sleep condition get **4hrs** of sleep a night. We are looking to *validate* this by analysing data collected in a **sample of 20 adults**, with a **Type 1 error rate of 0.05**.

Notation: Data collected from sample stored in variable `obs`, given population mean stored in variable `mu.given`, Type 1 error rate we are 'okay' with stored in `alpha`

We want to do a **two-tailed test**.

To conduct the One sample t-test in R:

```
t.test(obs, alternative = "two.sided",  
       mu = mu.given, paired = FALSE, var.equal = FALSE,  
       conf.level = 1 - alpha)
```



How to use the t.test function: **Two Sample t-test**

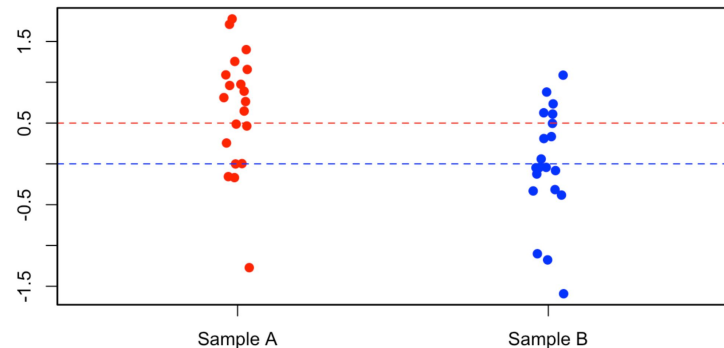
Example: We want to study if socio-economically disadvantaged people are more prone to addiction as compared to socio-economically 'stable' people. Let **Type 1 error rate be 0.01**.

Notation: Sample for disadvantaged group stored in variable `dis`, Sample for stable group stored in `st`. The mean difference in the null hypothesis is 0, Type 1 error Rate of 0.01 stored in `alpha`

Levene's test turned out to be not significant $\Rightarrow$ Equal variance across samples

To conduct the Two-Sample t-test:

```
t.test(x = dis, y = st, alternative = "greater",  
      mu = 0, paired = FALSE, var.equal = TRUE,  
      conf.level = 1 - alpha)
```



How to use the t.test function: **Paired t-test**

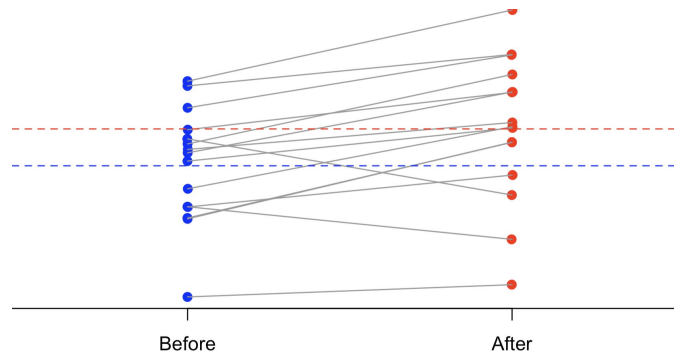
Example: Is there a *significant difference in growth rates* **before** and **after** the usage of growth supplements, with a **Type 1 error rate of 0.05**?

Notation: Sample growth rates before supplement use stored in variable `bef`, Sample growth rates after supplement use stored in variable `aft`. The mean difference in the null hypothesis is 0.

Say we want to do a **two-tailed test**.

To conduct the Paired t-test:

```
t.test(x = bef, y = aft, alternative = "two.sided",  
      mu = 0, paired = TRUE,  
      conf.level = 1 - alpha)
```



A Walkthrough on R

An Example from the Literature: Paired t-test

Figure 4b) from the following paper:

Liu, Q., Zhao, Y., Attanti, S. et al. Midbrain signaling of identity prediction errors depends on orbitofrontal cortex networks. *Nat Commun* 15, 1704 (2024).

<https://doi.org/10.1038/s41467-024-45880-1>

If you want to reproduce the analysis, GitHub Code:

https://github.com/QingfangLiu/OFC_midbrain_TMS/blob/main/SourceCode/Behavior.R

Note: The goal is to introduce the use of proxies (and not get lost in the math)

An Example from the Literature: Paired t-test

Context:

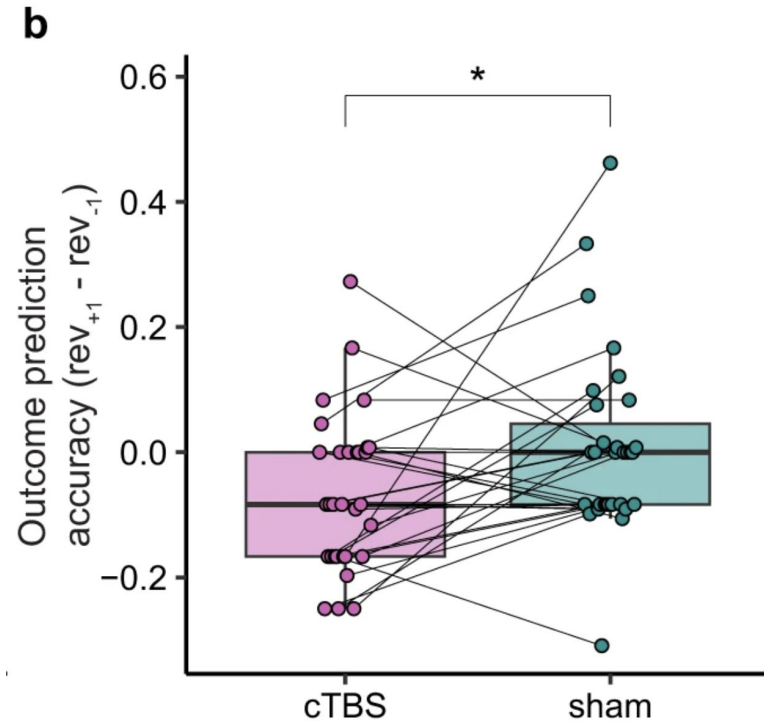
- Reversal Learning: People were made to perform a task, and their accuracy was noted. Then, they were asked to perform the ‘reverse’ of the task, and their accuracy was measured
 - As you’d expect, the accuracy would drop after reversal
- CTBS is a non-invasive brain stimulation technique that treats psychiatric and neurological conditions

So the question is, ‘Does Reversal learning ability decrease with CTBS’?

An Example from the Literature: Paired t-test

The question is, ‘**Does Reversal learning ability decrease with CTBS?**’

- The proxy is the accuracy drop (accuracy after reversal – accuracy before reversal)
- The t-test compares if the mean accuracy drop after CTBS is lower than the mean accuracy drop before CTBS
- This difference was significant ($p < 0.05$).
What does this mean?



An Example from the Literature: Paired t-test

```
51 #####
52 # change of accuracy from pre-reversal to post-reversal trial
53 #####
54
55 Acc_post_sham = subset(SubRevAcc, TMS == 'sham' & RevLoc == 1)$Acc
56 Acc_post_cTBS = subset(SubRevAcc, TMS == 'cTBS' & RevLoc == 1)$Acc
57 Acc_pre_sham = subset(SubRevAcc, TMS == 'sham' & RevLoc == -1)$Acc
58 Acc_pre_cTBS = subset(SubRevAcc, TMS == 'cTBS' & RevLoc == -1)$Acc
59
73 # change of acc from pre to post reversal trials (interaction test)
74 change_sham = Acc_post_sham - Acc_pre_sham
75 change_cTBS = Acc_post_cTBS - Acc_pre_cTBS
76
77 t.test(change_sham, change_cTBS, alternative = 'g', paired = T)
78 # t = 2.3442, df = 30, p-value = 0.01294
79
```

What we have learnt
is not useless!

Thank You!

Do reach out if you have any questions!